Figure 9.2: 5-Pillar Global Regulatory Compliance Shield: Statutory Banking Alignment

Formal Mapping: C-STGB Neuro-Symbolic Graph Invariants & Conformal Risk Control → International Supervisory Directives

COMPLIANCE AUDIT: 5 / 5 CERTIFIED

FATF 40 Rec.
GLOBAL MANDATE

Rec. 16 (Travel Rule) & Rec. 20

Continuous Entity Tracing

Originator/beneficiary identification and multi-hop peeling chain surveillance across cross-border banking wires.

C-STGB Architectural Solution

- **12-D Mass-Flow Invariants $\mathbf** {z}_{\mathrm{inv}}$
- **Kirchhoff conservation $\Phi_{\mat** hrm{flow}} \approx 1.0$
- **Tri-band harmonic attention $w(\De** lta t)$
- **90-day dormant chain persistence**

STATUTORY COMPLIANCE STATUS

VERIFIED: 100% Flow Traceability

FinCEN BSA
UNITED STATES

31 U.S.C. 5318(g) & Form 111

Autonomous SAR Dossiers

Mandatory detection of sub-$10k smurfing b ursts and automated filing of Suspicious Ac tivity Reports (SAR).

C-STGB Architectural Solution

- **Self-exciting Hawkes intensity $\l** ambda(t)$
- **Edge-trust filtering (**65.9% pru ned)
- **4-agent autonomous XML compiler**
- **Deterministic AST anti-hallucination**

STATUTORY COMPLIANCE STATUS

VERIFIED: 0% Entity Hallucination

Fed SR 11-7
US SUPERVISORY

OCC 2011-12 & SR 21-8 Guidance

Model Risk Governance

Comprehensive model validation, finite-sam ple risk bounds, drift stability, and tampe r-evident audit ledgers.

C-STGB Architectural Solution

- **Class-conditional CRC $\Gamma(u) ** subseteq \{0, 1\}$
- **Finite-sample DKW bounds ($1-\alph** a \ge 99\%)$
- **Two-sample KS drift test ($p = 0.4** 2$)
- **Append-only SHA-256 Merkle WORM**

STATUTORY COMPLIANCE STATUS

VERIFIED: Distribution-Free Risk Bound

EU 6AMLD / AI Act
EUROPEAN UNION

Arts. 13-15 & GDPR Art. 22

XAI & Corporate Liability

Explainability of automated triage, right to human explanation, and strict criminal l iability for institutional neglect.

C-STGB Architectural Solution

- **Tier-2 selective abstention $\Gamm** a(u) = \bot$
- **Human-in-the-loop analyst routing**
- **GNNExplainer subgraph masks**
- **Grammar verified decision lineage**

STATUTORY COMPLIANCE STATUS

VERIFIED: Human-in-the-Loop XAI

BFIU Circular 26
ASIA / EMERGING

MFS Master Circular & MLPA

Sub-Second Wire Clearance

Real-time Anti-Money Laundering surveillan ce across high-velocity mobile financial se rvices (MFS) agent networks.

C-STGB Architectural Solution

- **Sub-millisecond Fast-Path ($0.45\,** \text{ms}$)
- **Zero-copy PyArrow shared memory IPC**
- **Embedded DuckDB 2-hop cache (88.4%)**
- **Peak burst capacity: $31,250\,\tex** t{tx/s}$

STATUTORY COMPLIANCE STATUS

VERIFIED: 22.2x Faster than SLA

GLOBAL JURISDICTION COVERAGE

5 Statutory Frameworks

FATF, FinCEN BSA, Fed SR 11-7, EU 6AMLD, BFIU

MATHEMATICAL SAFETY GUARANTEE

≥99.0% Conformal Coverage

Finite-Sample Dvoretzky-Kiefer-Wolfowitz Bounds

EXPLAINABLE REGULATORY AI

0% Entity Hallucination

GNNExplainer Subgraph Masks + AST Grammar Verified

COURT-ADMISSIBLE EVIDENCE

FRE Rule 902(11) WORM

SHA-256 Merkle Proofs Sealed to Tamper-Proof Ledger